

Appendix C: IEEE 830 Template

Software Requirements Specification

for

Smart Parking System

Version 1.0

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Revision History

Name	Date	Reason for Changes	Version
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1. Introduction

1.1. Purpose

This document explains the specifics of the smart parking system. The purpose of this project is to create a smart parking system that allows users to locate, reserve, and pay for parking spaces efficient. It defines the system's functionalities, constraints, and interfaces for development and evaluation.

1.2. Document Conventions

This document follows the IEEE 830 Software Requirements Specification standard and uses consistent terminology and formatting conventions to ensure clarity and precision. All mandatory requirements are expressed using the keyword “**shall**”, indicating a mandatory system. Optional or non-mandatory features may use terms such as “may” or “can”.

Functional requirements are labeled using the format Req-x, where x represents a unique identifier. Non-functional requirements are labeled as NF-x and are categorized based on quality attributes such as performance, security, and reliability.

1.3. Intended Audience and Reading Suggestions

This document is intended for developers, testers who will test the system, stakeholders who want to know more about their investment, and users that want to know more about how the system works.

Developer: Developers who intend to work on the system whether to improve, modify, or extend the system must consult the document to understand the requirements and ensure consistency across all development phases.

Stakeholders: Stakeholders and including customers review this document to know if the requirements aligns with their requirements and goals. They use this specification to ensure that the system fulfills its intended purpose and delivers the required functionality.

Tester: The tester uses this document to test if the implemented system satisfies all specified requirements. It serves as a reference for designing test cases and ensuring that the system behavior matches the defined requirements.

1.4. Product Scope

The Smart Parking System is designed to assist drivers in locating, reserving, and paying for parking spaces in an efficient and manner. The system provides real-time information about parking availability and enables users to reserve spots in advance, reducing the time spent searching for parking. The system also supports administrative functionality for managing parking spaces, including adding, updating, and removing parking data. Additionally, it integrates with an external payment system to handle secure transactions.

1.5. References

No references

2. Overall Description

2.1. Product Perspective

The Smart Parking System is a standalone application that operates as an interface between users and parking infrastructure. It interacts with a centralized database to manage details like availability and reservations. It also has an external payment system to process transactions more securely. It may also receive data from sensors or manual input for availability of the parking spots.

System consists of three parts.

- User Interface
- Admin Interface for managing data
- Backend system that communicates that handles data processing, reservations, and communication with external services

2.2. Product Features

The Smart Parking System provides the following main features:

- **Parking Search:** Users can search for available parking spaces based on location.
- **Real-Time Availability:** The system displays up-to-date information about parking spot availability.
- **Parking Reservation:** Users can reserve available parking spaces in advance.
- **Payment Processing:** The system enables users to pay for parking reservations through an integrated payment service.
- **Reservation Management:** Users can view, modify, or cancel their reservations.
- **Parking Management (Admin):** Administrators can add, update, or remove parking spaces and manage system data.

2.3. User Classes and Characteristics

1.User(driver): They use the application to find and reserve parking space. Also, they can pay using the same application.

- Basic knowledge of how to use applications
- Frequent Usage
- Simple to use

2.Administrators: They are the ones responsible for managing the app and maintaining parking system by parking space data and monitoring system activity.

- Technical knowledge on how to use system.
- Responsible for system updates and accuracy.
- Requires access to admin functions.

3.Payment System: Payment system is an external service that processes user transactions securely.

- No direct user interaction
- Handling financial transactions
- Ensures secure and reliable payment processing

2.4. Operating Environment

The smart parking system is a web based or a mobile app that requires internet connection. The system can be accessed from modern browsers such as Chrome, Firefox etc. while also be used as an app that can be downloaded from android and IOS app stores. The backend system is expected to run on a server environment with database support (e.g., MySQL or similar database systems).

2.5. Design and Implementation Constraints

- The system must support real-time data processing to ensure accurate parking availability information. This requires efficient data handling and reliable communication between the system and the database.
- The system must ensure secure processing of user data and payment transactions. Appropriate encryption and authentication mechanisms must be implemented to protect sensitive information.
- If parking sensors are used, the system must be compatible with their data formats and communication protocols.
- The system must operate within typical hardware limitations of user devices, such as smartphones and personal computers.

2.6. User Documentation

This document is the only one.

2.7. Assumptions and Dependencies

The development of the parking system has several assumptions and external dependencies. Firstly, it is assumed that users have access to internet-enabled devices such as smartphones or computers to interact with the system. The system depends on having a network connection to provide real time information about parking spaces and requests. The system relies on accurate and up-to-date parking data which also needs either automatically to update itself or manually changed by admins. The system requires an external payment service for transactions. It is also assumed that the system database is properly maintained

3. System Features

3.1.1. Priority System

This feature allows user to get a parking space closer to their location and prioritize the closer ones for space. This way it will prioritize the nearby parking spots.

3.1.2. Stimulus/Response Sequences

Stimulus: User enters a location or opens their location and system finds it automatically.

Response: System retrieves and displays available parking spots.

3.1.3. Functional Requirements

Req-1: The system shall allow users to search for available parking spots.

Req-2: The system shall display available parking spots based on location.

Req-3: The system shall update parking availability in real time.

3.2 Reservation System

This feature allows user to get a parking space closer to their location and prioritize the closer ones for space. This way it will not show irrelevant spaces in other places

3.2.2 Stimulus/Response Sequences

Stimulus: User selects an available parking spot and reserves it.

Response: System reserves the parking spot and confirms the reservation.

3.2.3 Functional Requirements

Req-4: The system shall allow users to reserve a parking spot.

Req-5: The system shall prevent multiple users from reserving the same parking spot at the same time.

Req-6: The system shall allow user to cancel reservation.

Req-7: The system shall update the parking availability after a reservation is made or cancelled.

3.3 Payment Processing

This feature allows users to complete payments for reserved parking spots through an external payment system.

3.3.2 Stimulus/Response Sequences

Stimulus: User initiates payment for a reservation.

Response: System processes the payment and returns transaction result.

3.3.3 Functional Requirements

Req-8: The system shall allow users to make payments for reservations.

Req-9: The system shall securely transmit payment information to the external payment system.

Req-10: The system shall confirm successful payment transactions to the users.

Req-11: The system shall notify if the users payment fails.

3.4 Parking Management System

This feature allows administrators to manage parking spaces and system data, including adding, updating, and removing parking spots.

3.4.2 Stimulus/Response Sequences

Stimulus: Admin performs an action such as adding, updating or deleting the parking data.

Response: System updates the database.

3.4.3 Functional Requirements

Req-12: The system shall allow administrators to add new parking spots.

Req-13: The system shall allow admin to update the parking spot information.

Req-14: The system shall allow administrators to remove a parking spot.

Req-15: The system shall update parking availability after admin changes.

4. External Interface Requirements

4.1. User Interfaces Overview

The Smart Parking System provides a graphical user interface (GUI) accessible through web browsers and mobile devices. The interface is designed to be simple, intuitive, and efficient, allowing users to complete tasks with minimal effort.

From the user's perspective, the interaction with the system follows a clear sequence of screens:

Home Page: The user enters a location or allows a location access. The system displays available parking spaces in a map or list.

Selection Screen: The user selects a parking spot to view details such as availability and pricing.

Reservation page: The user confirms the reservation by selecting time and proceeding to the next step.

Payment page: The user enters payment details and confirms the transaction.

Confirmation Screen: The system displays a success or failure message based on the payment result.

Interface Requirements

The interface must meet the following minimum requirements:

- The system shall provide a graphical interface with clear navigation and layout.
- Standard interface elements such as buttons (e.g., "Search", "Reserve", "Pay") must be consistently displayed.
- Each screen shall provide clear instructions and feedback messages.
- Error messages shall be displayed when invalid input or system failures occur.
- The interface shall support responsive design for both desktop and mobile devices.
- Basic usability features such as clear labels and simple navigation must be implemented.

4.2. Hardware Interfaces

The Smart Parking System interfaces with user devices and optional parking hardware components.

Users access the system through devices such as smartphones, tablets, and desktop computers. These devices provide input through touchscreens, keyboards, or pointing devices, and display system output through graphical interfaces. The system communicates with these devices via standard internet protocols.

The system may also interface with parking sensors installed in parking areas. These sensors detect the occupancy status of parking spaces and transmit data indicating whether a spot is available or occupied. This data is sent to the system which processes it to update parking availability in real time.

The communication between the system and hardware components is expected to occur over standard communication protocols.

The interaction between the system and hardware components is limited to receiving status data from sensors and sending updated availability information to user devices. No direct control over hardware components is required for basic system functionality.

4.3. Software Interfaces

The Smart Parking System interacts with several external software components to provide its core functionality.

The system interfaces with a database management system to store and retrieve data related to users, parking spaces, reservations, and transactions. The database is responsible for maintaining data consistency and supporting real-time updates of parking availability.

The system also interfaces with an external payment system to process financial transactions. When a user initiates a payment, the system securely transmits payment details to the payment service and receives a response indicating whether the transaction was successful or failed.

Communication between the Smart Parking System and external software components is performed using standard web-based protocols. Data exchanged between systems includes user information, reservation details, parking availability data, and payment transaction results.

5. System Features/Modules

5.1. Parking Search

5.1.1 Description and Priority

This feature allows user to get a parking space closer to their location and prioritize the closer ones for space. This way it will prioritize the nearby parking spots.

5.1.2 Stimulus/Response Sequences

Stimulus: User enters a location or opens their location and system finds it automatically.

Response: System retrieves and displays available parking spots.

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Req-1: The system shall allow users to search for available parking spots.

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This feature allows user to get a parking space closer to their location and prioritize the closer ones for space. This way it will not show irrelevant spaces in other places

5.2.2 Stimulus/Response Sequences

Stimulus: User selects an available parking spot and reserves it.

Response: System reserves the parking spot and confirms the reservation.

5.2.3 Functional Requirements

Req-4: The system shall allow users to reserve a parking spot.

Req-5: The system shall prevent multiple users from reserving the same parking spot at the same time.

Req-6: The system shall allow user to cancel reservation.

Req-7: The system shall update the parking availability after a reservation is made or cancelled.

5.3 Payment Processing

This feature allows users to complete payments for reserved parking spots through an external payment system.

5.3.2 Stimulus/Response Sequences

Stimulus: User initiates payment for a reservation.

Response: System processes the payment and returns transaction result.

5.3.3 Functional Requirements

Req-8: The system shall allow users to make payments for reservations.

Req-9: The system shall securely transmit payment information to the external payment system.

Req-10: The system shall confirm successful payment transactions to the users.

Req-11: The system shall notify if the users payment fails.

5.4 Parking Management System

This feature allows administrators to manage parking spaces and system data, including adding, updating, and removing parking spots.

5.4.2 Stimulus/Response Sequences

Stimulus: Admin performs an action such as adding, updating or deleting the parking data.

Response: System updates the database.

5.4.3 Functional Requirements

Req-12: The system shall allow administrators to add new parking spots.

Req-13: The system shall allow admin to update the parking spot information.

Req-14: The system shall allow administrators to remove a parking spot.

Req-15: The system shall update parking availability after admin changes.

6. Nonfunctional Requirements

6.1. Performance

NF-1.1: The system shall respond to user requests within 2 seconds under normal circumstances.

NF-1.2: The system shall update parking information in real time.

6.2. Security

NF-2.1: The system shall securely transmit the payment data.

NF-2.2: The system shall protect user data from unauthorized access.

NF-2.3: The system shall ensure secure communication with an external payment system.

6.3. Usability Requirements

NF-3.1: The system shall provide a simple and intuitive user interface.

NF-3.2: The system shall require the user to be able to complete basic tasks without prior knowledge.

6.4. Reliability Requirements

NF-4.1: The system shall maintain at least %99 uptime.

NF-4.2: The system shall handle errors without data loss.